

Question

Sodium azide is a commonly used reagent in organic chemistry. In the laboratory, the purity of sodium azide is often determined by titration analysis, and the specific steps are as follows:

1. Accurately weigh 0.4620 g of dried arsenic trioxide into a 250 mL beaker, add 100 mL of $100 \text{ g} \cdot \text{L}^{-1}$ sodium hydroxide solution to dissolve it, then transfer to a 1 L volumetric flask. Add 250 mL of water and 250 mL of $3 \text{ mol} \cdot \text{L}^{-1}$ sulfuric acid, and dilute to the mark.
2. Precisely pipette 50.00 mL of arsenic trioxide standard solution into a 500 mL conical flask, add 3 drops of $2.5 \text{ g} \cdot \text{L}^{-1}$ osmium tetroxide solution, then add 25.00 mL of the ceric ammonium nitrate solution to be standardized. Dilute with water to 150 mL, add 30 mL of $3 \text{ mol} \cdot \text{L}^{-1}$ sulfuric acid and 2 drops of ferroin indicator, and titrate with the ferrous ammonium sulfate solution to be standardized until the endpoint is reached. The titration consumes 39.90 mL of ferrous ammonium sulfate standard solution.
3. Using all the reagents mentioned above except arsenic trioxide, the titration consumes 49.18 mL of ferrous ammonium sulfate standard solution.
4. Accurately weigh 2.7233 g of sodium azide sample, dissolve it in water, and dilute to 250 mL. Pipette 20.00 mL of this solution into a 500 mL conical flask, add 40.00 mL of ceric ammonium nitrate standard solution, and allow the reaction to proceed in a cold water bath. After the reaction is complete, dilute to 150 mL, add 30 mL $3 \text{ mol} \cdot \text{L}^{-1}$ sulfuric acid and 2 drops of ferroin indicator, then titrate with ferrous ammonium sulfate standard solution until the endpoint is reached. The titration consumes 15.32 mL of ferrous ammonium sulfate standard solution.

Regarding this analytical process, the following statements are made:

- A. Dried arsenic trioxide can be directly used to prepare the standard test solution.
- B. The osmium tetroxide added undergoes a change in oxidation state during the analysis.
- C. The ionic equation for the chemical reaction occurring after adding ceric ammonium nitrate in step 2, when balanced with the simplest integer coefficients, has a sum of product coefficients equal to 5.

D. The concentration of the prepared arsenic trioxide solution is 0.002335 mol/L.

E. The concentration of the ceric ammonium nitrate solution is 0.09900 mol/L.

F. The concentration of ferrous ammonium sulfate is 0.04032 mol/L.

G. The content of sodium azide is 95.16 %.

Among these statements, the number of correct ones is (for calculation-related options, an error less than 1 % is considered correct).

A. 0

B. 1

C. 2

D. 3

E. 4

F. 5

G. 6

H. 7

Answer

Correct Answer: **G**

Detailed Explanation

Arsenic trioxide is a primary standard substance with strong stability. After drying, it can be used to directly prepare standard solutions, so statement A is correct.

CHECKPOINT

1 PTS

Arsenic trioxide is a primary standard substance, so statement A is correct

The added osmium tetroxide acts as a catalyst in the analysis process, catalyzing the oxidation reaction. Osmium tetroxide oxidizes arsenic trioxide and is reduced to the hexavalent state itself. Ammonium cerium(IV) nitrate then oxidizes hexavalent osmium to osmium tetroxide, forming a catalytic cycle with a change in oxidation state, so statement B is correct.

CHECKPOINT

1 PTS

Osmium tetroxide acts as a catalyst to form an oxidation cycle, so statement B is correct

In the analysis process, the main purpose of process 1 is to prepare a standard solution of arsenic trioxide, which dissolves in sodium hydroxide to produce arsenite ions AsO_3^{3-} , and then is acidified to arsenous acid H_3AsO_3 . The concentration of the arsenic trioxide standard solution can be calculated as follows:

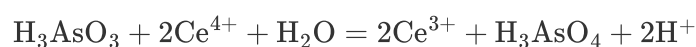
$$c(\text{As}) = n/V = 0.4620/197.84/1 = 0.002335 \text{ mol/L}, \text{ so statement D is correct.}$$

CHECKPOINT

1 PTS

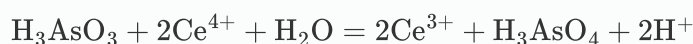
$c(\text{As}) = 0.002335 \text{ mol/L}$, so statement D is correct.

Processes 2 and 3 are used to simultaneously standardize ferrous ammonium sulfate and ammonium cerium(IV) nitrate solutions. Ammonium cerium(IV) nitrate is used to oxidize arsenic trioxide, and the remaining ammonium cerium(IV) nitrate is back-titrated with ferrous ammonium sulfate solution; at the same time, a blank experiment without arsenic trioxide is used to determine the concentration of ferrous ammonium sulfate. The reactions that occur during the process are:



CHECKPOINT

1 PTS



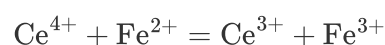
The sum of the coefficients of the products of this reaction is 5, so statement C is correct.

CHECKPOINT

0.5 PTS

The sum of the coefficients of the products of this reaction is 5, so statement C is correct.

The back-titration equation is



The difference in the amount of ferrous ammonium sulfate used in processes 2 and 3 corresponds to the amount of ammonium cerium(IV) nitrate that oxidized the arsenic trioxide; the amount of arsenic trioxide added is

CHECKPOINT

1 PTS

The difference in the amount of ferrous ammonium sulfate used corresponds to the amount of ammonium cerium(IV) nitrate that oxidized the arsenic trioxide

$$n = cV = 0.002335 \times 0.0500 = 0.00011675 \text{ mol}$$

According to the redox stoichiometric relationship, $c(\text{Fe}^{2+}) - c(\text{Ce}^{4+}) - 2c(\text{H}_3\text{AsO}_3) - 4c(\text{As}_2\text{O}_3)$

The concentration of ferrous ammonium sulfate is obtained as

$$c(\text{Fe}) = n/V = \frac{4 \times 0.00011675}{(49.18 - 39.90) \times 0.001} = 0.05032 \text{ mol/L}$$

CHECKPOINT

1 PTS

$c(\text{Fe}) = 0.05032 \text{ mol/L}$, so statement F is incorrect

Therefore, statement F is incorrect.

Through the blank experiment, all the ferrous ammonium sulfate in process 3 is used to reduce ammonium cerium(IV) nitrate, from which the concentration of ammonium cerium(IV) nitrate can be deduced:

$$c(\text{Ce}) = c(\text{Fe}) \frac{V(\text{Fe})}{V(\text{Ce})} = 0.05032 \times \frac{49.18}{25.00} = 0.09895 \text{ mol/L}$$

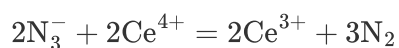
Therefore, statement E is correct.

CHECKPOINT

1 PTS

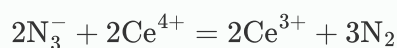
$c(\text{Ce}) = 0.09895 \text{ mol/L}$ Statement E is correct.

Process 4 uses ammonium cerium(IV) nitrate to titrate sodium azide, and then ferrous ammonium sulfate is used to back-titrate the ammonium cerium(IV) nitrate. The equation for the first step of oxidizing sodium azide is



CHECKPOINT

1 PTS



According to the reaction stoichiometry, one molecule of sodium azide corresponds to one molecule of ammonium cerium(IV) nitrate, so the amount of sodium azide is calculated as

$$n(\text{N}_3^-) = n(\text{Ce}_{\text{react}}) = n(\text{Ce}_{\text{total}}) - n(\text{Fe}) = 0.09895 \times 0.04 - 0.05030 \times 0.01532 = 0.0031874 \text{ mol}$$

$$\omega(\text{N}_3^-) = m(\text{N}_3^-)/m(\text{total}) = 0.0031874 \times 65.02 \times (250/20)/2.7233 = 0.9513$$

CHECKPOINT

1 PTS

$\omega(\text{N}_3^-) = 0.9513$, so statement G is correct

That is, the content is 95.13 %, statement G is correct.

In summary, there are six correct statements, so choose option G.